DATA SCIENCE ALLIANCE • EXECUTIVE SYNTHESIS BRIEFING

Downtown San Diego Unsheltered Homelessness: Strategic Synthesis & Policy Roadmap (2017–2026)

Longitudinal Econometrics, Spatial Panel Dynamics, Machine Learning Forecasting & Regional Policy Impacts

PARTNERSHIP & AUTHORS

Data Science Alliance (DSA) in partnership with DSDP & RTFH
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DATA FOUNDATION

9 Years of Monthly Field Counts (Jan 2017–Dec 2025, 108 months, 104 published)
Point-in-Time (PIT) Regional Censuses (2017–2025)

ANALYTICAL FRAMEWORK

Getis-Ord Gi* • 5-Fold Walk-Forward Cross-Validation • Policy Regime OLS • The 4 Public Communication Rules

EXECUTIVE FRAMING & CORE FINDINGS

Executive KPI Dashboard: Crisis Peak, Enforcement Contraction & Hotspot Migration

2,103

CRISIS PEAK (MAY 2023)

Highest headline count in downtown history (2,103.3); culmination of 3-year surge.

851

RECENT COUNT (JAN 2025)

-59.5% reduction from peak; Dec 2025 count reached 985.4.

54.0%

EAST VILLAGE VOLUME

Epicenter of downtown homelessness (historical monthly average 552.0 individuals).

120 m

HOTSPOT MIGRATION VECTOR

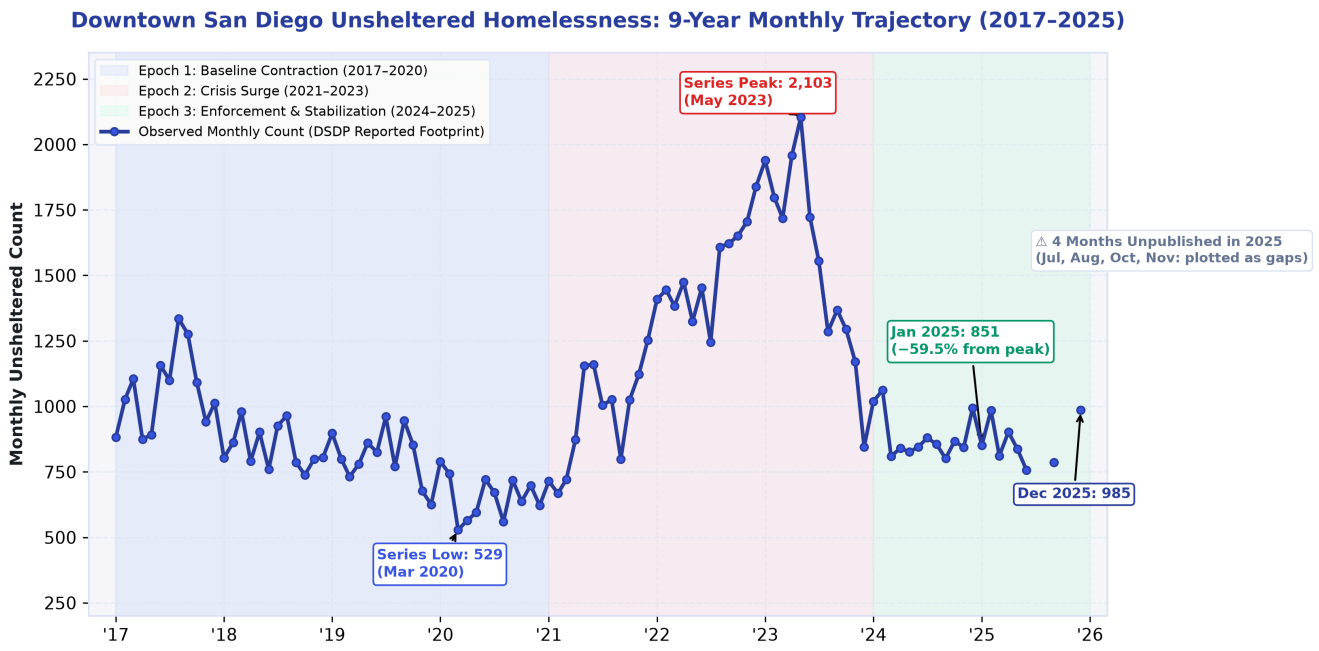
Cluster centroid shifted 120.4 m bearing 160° (SSE) toward South East Village.

THE STRATEGIC REALITY: DOWNTOWN CONTRACTION VS. REGIONAL DISPLACEMENT

- **Three Trajectory Epochs:** Downtown transitioned through Baseline Contraction (2017–2020: annual mean 1,058 → 654), Crisis Surge (2021–2023: annual mean 960 → 1,563, peak 2,103 in May 2023), and Enforcement Pullback (2024–2025: annual mean ~887 and 864, Jan 2025 count 851).
- **Debunking Seasonality:** Non-parametric tests prove 8 of 9 downtown series have NO statistically significant seasonality (Overall series H=2.66, p=0.995). Resource budgeting based on seasonal surges is empirically unsupported.
- **The Boundary Expansion Wedge:** In 2022, DSDP added 121 peripheral blocks, accounting for 3.7% to 32.8% of reported counts. Longitudinal comparisons require the balanced 261-block spatial panel (+71% in 2022).
- **Regional Divergence:** Downtown's share of county unsheltered fell from 37.5% (2023) to 14.9% (2025), yet countywide unsheltered reached 6,110 (2024, series high) and 5,714 (2025). Divergence indicates spatial redistribution rather than resolved homelessness.

LONGITUDINAL DYNAMICS

The 9-Year Trajectory: Three Distinct Macro Policy Epochs (2017–2025)



Epoch 1: Baseline Contraction (2017–2020)

Low of 529 in March 2020 (-38.2% drop 2017→2020)

Pre-COVID contraction and initial shelter interventions; series low of 529 observed at COVID-19 onset in March 2020.

Epoch 2: The Crisis Surge (2021–2023)

May 2023 Peak of 2,103 (+297% from March 2020 low)

Multi-year escalation driven by shelter de-densification, economic stress, and post-moratoria housing vulnerability; annual mean reached 1,563 in 2023.

Epoch 3: Enforcement & Stabilization (2024–2025)

Jan 2025 Count of 851 (-59.5% from peak)

Sharp contraction following the July 2023 Unsafe Camping Ordinance and Safe Sleeping site rollouts; annual counts stabilized near ~887 (2024) and 864 (2025). 4 months unpublished in 2025.

ECONOMETRIC SEASONALITY PROOF

Debunking the Seasonality Myth: Homelessness Is Not Driven by Weather

Kruskal-Wallis Non-Parametric Tests (df=11)

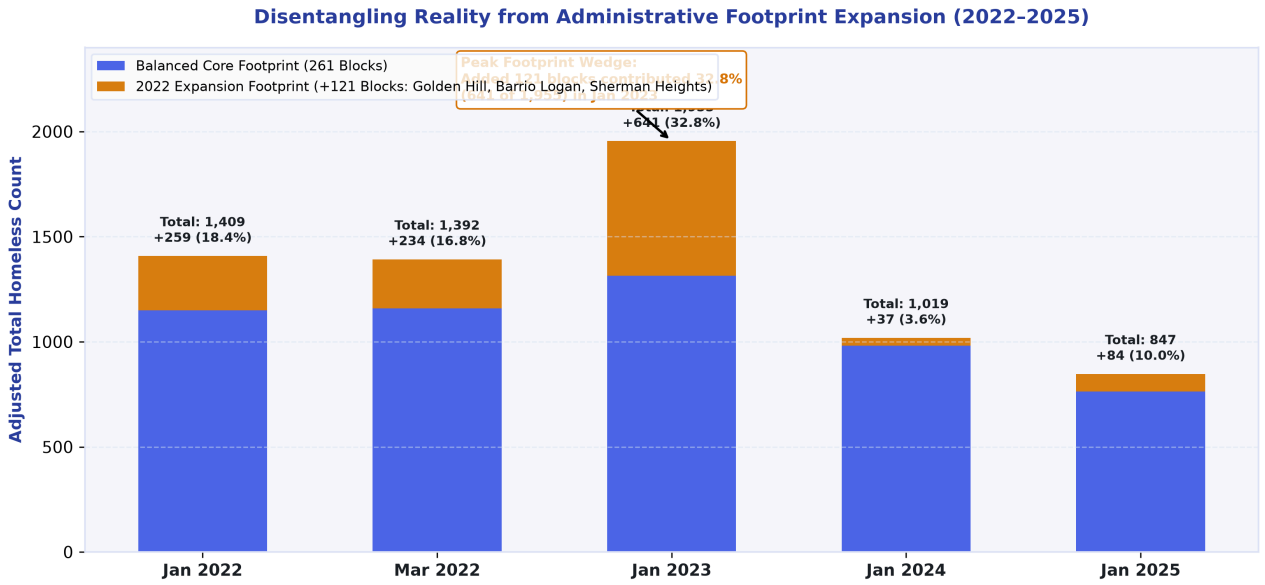
Geographic Series	KW H	KW p	ANOVA p	Verdict
Overall — Reported Footprint	2.66	0.995	0.998	Zero Seasonality (p > 0.05)
Overall — Fixed Core Six	9.00	0.622	0.670	Zero Seasonality (p > 0.05)
East Village (Epicenter)	14.16	0.224	0.219	Zero Seasonality (p > 0.05)
City Center (Core)	4.86	0.938	0.913	Zero Seasonality (p > 0.05)
Gaslamp Quarter	14.57	0.203	0.087	Zero Seasonality (p > 0.05)
Columbia	8.76	0.644	0.473	Zero Seasonality (p > 0.05)
Cortez	12.73	0.312	0.047	Mixed (ANOVA sig, KW non-sig)
Marina	19.88	0.047	0.162	Mixed (KW sig, ANOVA non-sig)
Outside Perimeter	11.45	0.406	0.440	Zero Seasonality (p > 0.05)

WHY THIS MATTERS FOR MUNICIPAL BUDGETING

- **The 'Weather Surge' Fallacy:** Civic planning often assumes winter drops or summer surges. In reality, calendar months explain virtually none of monthly count variation across downtown (p = 0.995).
- **Macro Drivers Dominate:** Multi-year shifts, shelter capacity adjustments, and regional housing dynamics drive changes—not temperature or rainfall.
- **Eliminate Fluctuating Allocations:** Emergency seasonal shelter beds fail to solve baseline need. Funding must be committed to permanent year-round operating capacity.
- **Borderline Disagreements:** KW and ANOVA disagree for Marina (KW p=0.047, ANOVA p=0.162) and Cortez (KW p=0.312, ANOVA p=0.047); 8 of 9 series are strictly non-significant under the primary KW test.

SPATIAL PANEL ACCOUNTING

The Boundary Expansion Wedge: Disentangling Reality from Map Changes



ACCOUNTING FOR THE 121-BLOCK EXPANSION

- **The Administrative Shift:** In January 2022, DSDP expanded survey coverage from 261 blocks to 382 blocks by incorporating 121 peripheral blocks across Golden Hill (52), Barrio Logan (35), and Sherman Heights (34).
- **Quantifying the Wedge:** Between 3.7% and 32.8% of reported post-2021 count totals came from the added blocks. At the January 2023 peak, 640.6 of 1,955.1 individuals (32.8%) were in the expansion area.
- **Underlying Crisis Remained Real:** On the balanced 261-block panel, counts still surged +71% in 2022. The crisis was real, but unadjusted comparisons conflate geography with population changes.
- **Data Engineering Fix:** Enforced strict longitudinal panel balance (`BlockPanel_Balanced_261.csv`) across all 12 digitized count dates (2018–2025).

GEOGRAPHIC CONCENTRATION

Where Homelessness Concentrates: The East Village Epicenter

Top Persistent Hotspots (Getis-Ord Gi*)

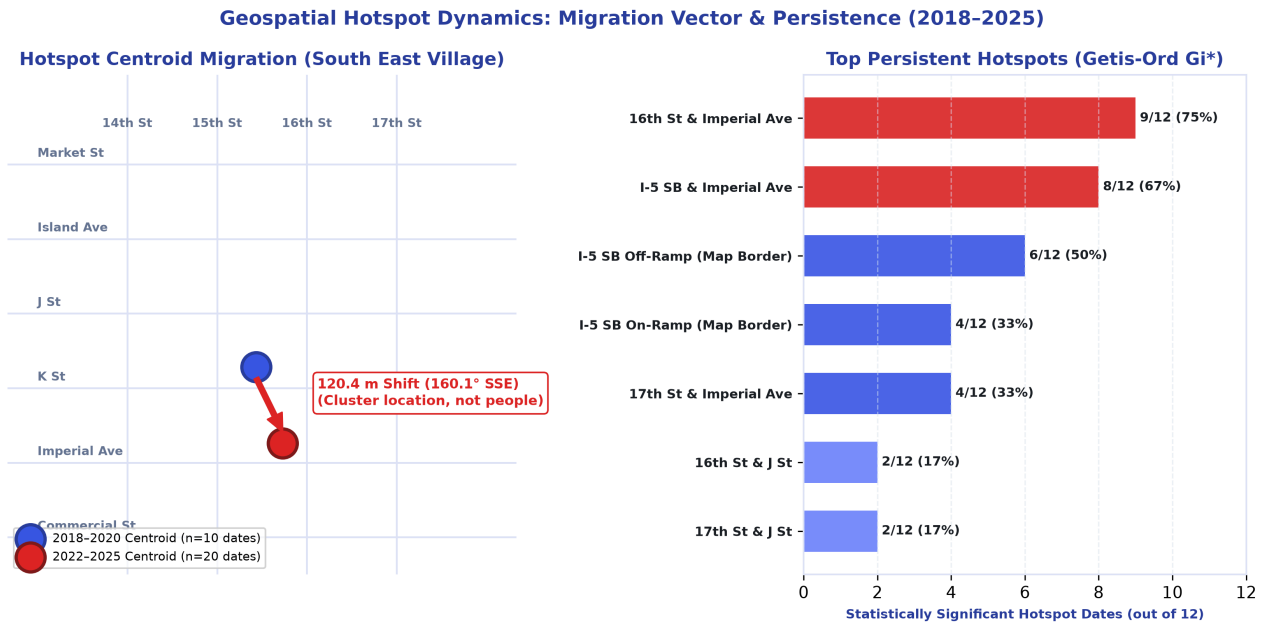
Rank	Block Identifier	Area	Hotspot Dates	Rate	Mean Count
#1	16th St & Imperial Ave	South East Village	9 / 12 dates	75%	5.9
#2	I-5 SB & Imperial Ave	South East Village	8 / 12 dates	67%	4.4
#3	I-5 SB Off-Ramp (Map Border)	South East Village	6 / 12 dates	50%	6.8
#4	I-5 SB On-Ramp (Map Border)	South East Village	4 / 12 dates	33%	1.3
#5	17th St & Imperial Ave	South East Village	4 / 12 dates	33%	5.5
#6	16th St & J St	South East Village	2 / 12 dates	17%	13.3
#7	17th St & J St	South East Village	2 / 12 dates	17%	12.2

THE BUILT-ENVIRONMENT MAGNET

- **East Village Volume Dominance:** Historically accounts for 54.0% of all downtown homelessness (mean 552.0 individuals). South East Village alone accounts for over 317 individuals/month.
- **100% of Top 10 in South East Village:** Every single one of the top 10 persistent hotspot blocks is located in South East Village, forming a durable geographic cluster.
- **Built-Environment Infrastructure:** Hotspots cluster around major infrastructure anchors: the 12th & Imperial Transit Center, social service facilities, and Caltrans freeway underpasses providing physical overhead cover.

CLUSTER MIGRATION VECTOR

Geospatial Hotspot Dynamics: The 120-Meter Southward Displacement Vector



DRIVERS OF CLUSTER DISPLACEMENT

- **120.4-Meter Migration Vector:** Between early (2018–2020, n=10) and recent (2022–2025, n=20) periods, the count-weighted hotspot centroid shifted 120.4 meters on a bearing of 160.1° (south-south-east).
- **Unweighted Centroid:** The unweighted centroid moved 62 meters, confirming the southern pull is driven by both cluster location and density.
- **Northern Enforcement & Sidewalk Access:** Increased enforcement of sidewalk clearance in core commercial areas (Gaslamp, Core/City Center) pushed unhoused clusters southward.
- **Cluster Location vs Individual Tracking:** This vector reflects where statistically significant hotspot clusters sit across count dates, not direct tracking of individual people.

PREDICTIVE INTELLIGENCE

Multi-Model Machine Learning Benchmark: 5-Fold Walk-Forward Cross-Validation

5-Fold Walk-Forward CV Performance (Table 8.1, Zero Leakage)

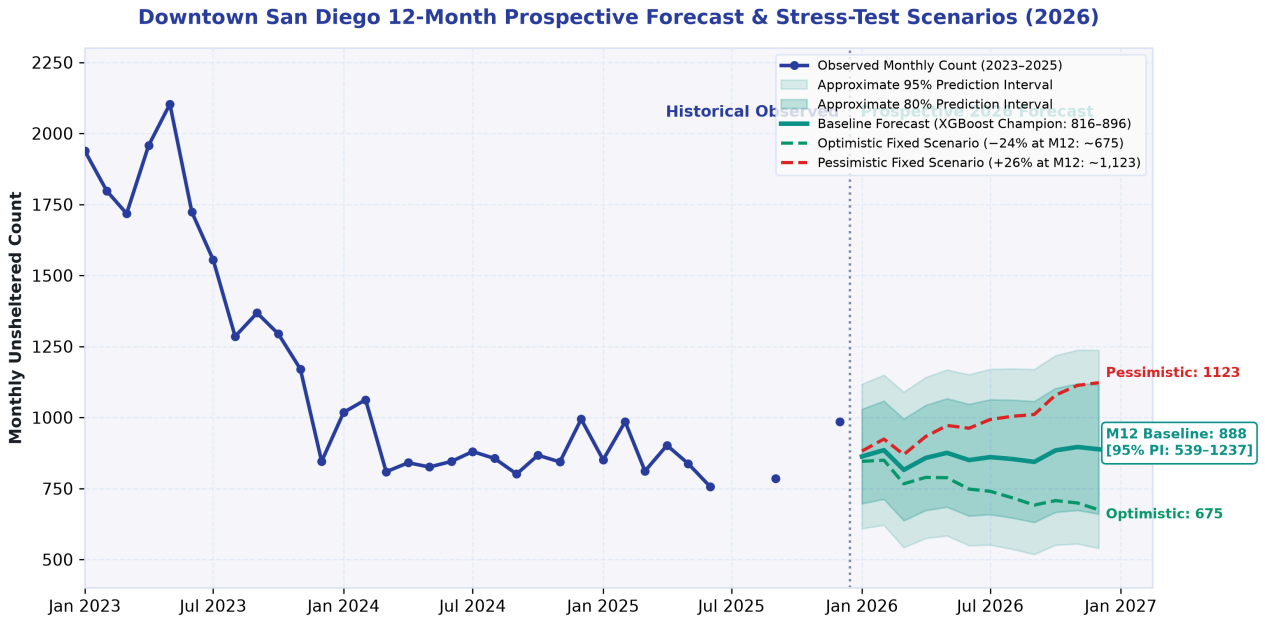
Model Architecture	1-Month MAPE	3-Month MAPE	6-Month MAPE	12-Month CV MAPE
SARIMA (ACF/PACF Grid)	5.4% ± 4.1%	14.3% ± 10.3%	24.7% ± 20.9%	39.6% ± 35.5%
Holt-Winters Exp Smooth	13.0% ± 9.2%	17.6% ± 10.4%	25.2% ± 18.2%	38.4% ± 28.5%
XGBoost (Champion)	14.0% ± 8.9%	19.9% ± 14.8%	26.7% ± 18.5%	**38.3% ± 26.0%**
Naive Seasonal (t-12)	39.5% ± 31.9%	40.1% ± 31.3%	48.5% ± 35.4%	51.7% ± 24.5%
Prophet (Changepoints)	42.9% ± 26.0%	42.7% ± 26.0%	51.3% ± 31.9%	64.2% ± 35.4%

CORE FORECASTING TAKEAWAYS

- **Short vs Long Horizons:** SARIMA dominates short horizons (1-month MAPE 5.4%). At 12 months, XGBoost (38.3%), Holt-Winters (38.4%), and SARIMA (39.6%) are a statistical tie given 26–36 point fold SD.
- **XGBoost Champion:** Retained as champion for the overall series due to lowest 12-month mean (38.3%) and strong hold-out performance (8.5% on 2025).
- **2025 Hold-Out Baseline Victory:** Naive seasonal achieved 6.3% on 2025 because 2025 repeated 2024 levels; highlights the importance of retaining naive baselines in all benchmarking.
- **Prophet Failure:** Prophet scored worst at all horizons (64.2% at 12 months) because fixed changepoints failed to detect the post-peak inflection.

PROSPECTIVE FORECASTING

2026 Forward Outlook: Stress-Testing Three Strategic Policy Scenarios

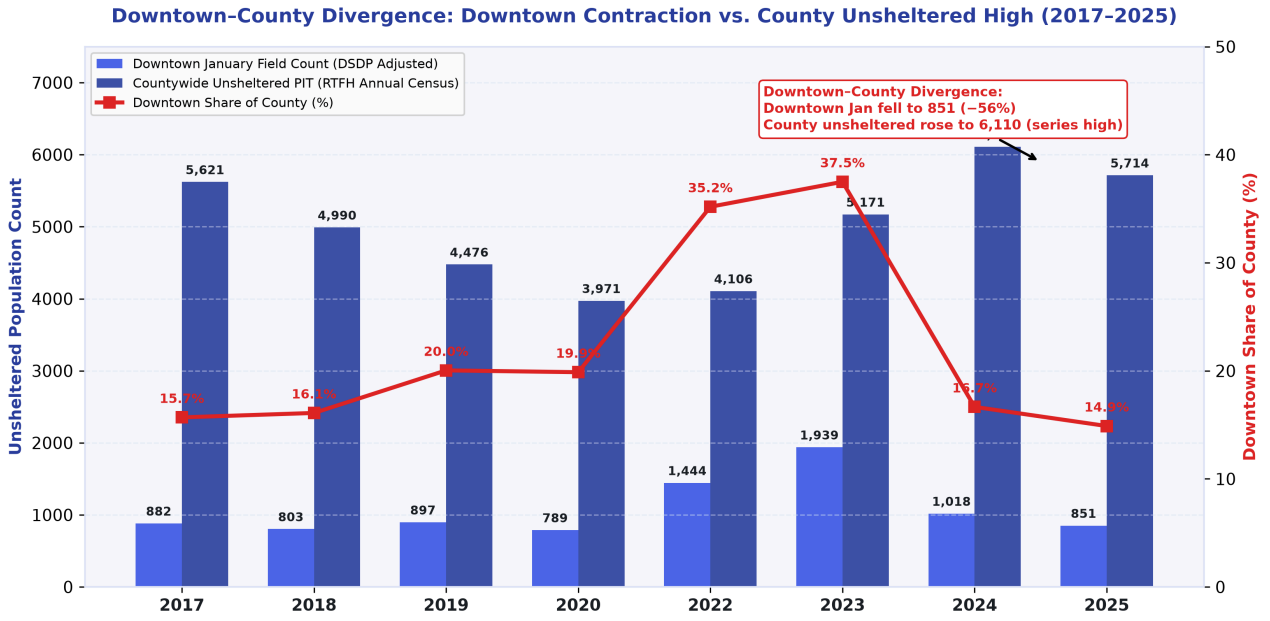


PROSPECTIVE 2026 SCENARIOS

- **Baseline Outlook:** Monthly counts projected to remain range-bound between 816 and 896 individuals (Month 12 Dec 2026: 888, approximate 95% PI 539–1,237).
- **Fixed What-If Scenarios:** Scenarios are fixed stress-test paths (-2% and +2.2% monthly), NOT model-based policy simulations.
- **Optimistic Path (-24% at M12):** Projects a reduction to ~675 individuals by Dec 2026 under favorable shelter turnover and intake.
- **Pessimistic Path (+26% at M12):** Projects an increase to ~1,123 individuals by Dec 2026 under shelter constraints and economic displacement.
- **Planning with Ranges:** Decision-makers must plan around calibrated uncertainty bands rather than single-point forecasts.

REGIONAL POLICY DYNAMICS

The Regional Displacement Paradox: Downtown Collapse vs. Countywide High



THE DISPLACEMENT REALITY

- **Downtown Share Plunged:** Downtown's share of countywide unsheltered homelessness plunged from 37.5% (Jan 2023) to 14.9% (Jan 2025).
- **County Unsheltered High:** Simultaneously, the RTFH Point-in-Time count recorded 6,110 unsheltered individuals in 2024 (series high) and 5,714 in 2025.
- **Spatial Displacement vs Resolution:** Downtown reductions coincided with regional increases, consistent with encampment displacement into riverbeds, canyons, and neighboring municipalities.
- **Empirical Limit:** The project lacks individual tracking data to prove where displaced persons moved; findings show macro divergence, not causal proof.

MACRO ECONOMETRICS & GOVERNANCE

Countywide Macro Estimation Model & Mandatory Communication Rules

POLICY REGIME OLS CHAMPION MODEL

- **Ratio Stability Rejected:** One-Way ANOVA rejects ratio stability across regimes (F = 66.51, p = 0.0002, eta^2 = 0.964). Pearson correlation is r = -0.061 over 8 paired years.
- **Econometric Equation:**
  `Y = 4,513.26 + 0.1859 * X + 1,225.04 * D_post`
  Where X = Downtown Jan count, D_post = 1 for 2024/2025.
- **Fit & Diagnostics:** R^2 = 0.504 | Adj R^2 = 0.306 | LOOCV RMSE = 815.3 | Bias = -0.3%. Neither slope (p=0.78) nor D_post (p=0.075) is statistically significant at 5%.
- **Current County Estimate:**
  5,900 [95% PI: 3,900 to 8,000 individuals]
  Margin of error is ±2,050 individuals (±35%) using Student's t distribution (df=5).

THE 4 MANDATORY PUBLIC COMMUNICATION RULES

- **Rule 1: Prohibit Raw Single-Point Precision:** Never publish uncalibrated single figures like '5,921'. False precision misleads the public and damages institutional credibility.
- **Rule 2: Mandate Rounding to Nearest 100:** Always round estimated point totals to the nearest 100 individuals (mandate '5,900' instead of '5,921').
- **Rule 3: Always Publish Calibrated 95% Interval:** Every headline figure MUST be accompanied by its calibrated Student's t prediction interval: *'5,900 [3,900 to 8,000]'*.
- **Rule 4: Mandatory Policy Regime Disclosure:** Public statements must explicitly disclose that the estimate derives from downtown counts and assumes the post-July 2023 Unsafe Camping Ordinance enforcement environment continues.

STRATEGIC RECOMMENDATIONS

Municipal Policy Action Roadmap: Strategic Recommendations (R1–R8)

IMMEDIATE ACTION (0–3 MONTHS)

Outreach Focus & Reporting Governance

- **R1. Focus on Hotspots:** Shortlist 16th–17th & Imperial and I-5 ramps for coordinated multidisciplinary outreach.
- **R2. Mandate Range Planning:** Enforce Rules 1–4 across all municipal and media releases; plan with ranges.
- **R3. Clarify Boundary Scope:** Always state whether figures cover the balanced 261-block core or expanded 382-block footprint.
- **Lead Agencies:** City Homelessness Strategies, DSDP, Service Outreach.

MEDIUM-TERM (6–12 MONTHS)

Data Integrity & Empirical Validation

- **R4. Complete Component Data:** Require DSDP to publish complete individual/tent/vehicle counts with consistent quadrant totals.
- **R5. Test Displacement Directly:** Launch coordinated cross-jurisdictional outreach tracking to verify regional movement.
- **R6. Annual Retraining:** Re-estimate forecasts and county model upon release of 2026 RTFH PIT results.
- **R7. Upgrade Shelter Data:** Replace synthetic placeholder shelter series with administrative Housing Inventory Count (HIC) data.

LONG-TERM (1–3 YEARS)

Spatial Digitization & System Infrastructure

- **R8. Digitize Remaining Maps:** Digitize the ~78 remaining monthly block maps to enable seasonal and post-ordinance spatial analysis.
- **Regional HMIS Integration:** Coordinate live bed availability tracking between city and county providers.
- **Longitudinal Evaluation:** Measure true regional outcomes rather than corridor-by-corridor displacement.
- **Lead Agencies:** County Health & Human Services, RTFH, DSA.

OPERATIONAL DECISION TOOL

Interactive Decision-Support System: The Streamlit Executive Dashboard

THE 7 SPECIALIZED DASHBOARD MODULES (TASK 9.1 & 10)

- **Tab 1: Historical Trends:** Longitudinal time series (2017–2025), multiplier regime markers, epoch shading, component toggles.
- **Tab 2: 12-Month Forecast:** Prospective 2026 projections, model selectors (XGBoost champion, HW, SARIMA, Prophet), 80%/95% fan envelopes, scenario stress tests.
- **Tab 3: Hotspot Surveillance:** Folium interactive choropleth, 12 count dates, 261 vs 382 panel toggle, Getis-Ord Gi* classifications (95%, 99%).
- **Tab 4: Neighborhood Deep-Dive:** 4-pane STL decomposition, monthly seasonal indices, Kruskal-Wallis non-parametric significance displays.
- **Tab 5: County Estimation Calculator:** Interactive Policy Regime OLS calculator enforcing The 4 Public Communication Rules with Student's t intervals.
- **Tab 6: Methodology & About:** Documentation of 4 multiplier schedules, 2022 expansion (+121 blocks), data audits and QA flags.
- **Tab 7: Anomaly Surveillance:** Shewhart control charts (±2σ, ±3σ) and CUSUM (k=0.5, h=4.0) early-warning surveillance engine for rapid operational alerts.

DEPLOYMENT & VERIFICATION INFO

- **Local Live URL:** `http://localhost:8501` (Fast interactive response time)
- **Cloud Production URL:** `https://downtown-sd-homelessness.streamlit.app`
- **Automated Quality:** 46 / 46 pytest unit tests passing with zero regressions.
- **Data Pipeline SLA:** `@st.cache_data` provides fast sub-second cached lookups across all 7 tabs.
- **Target Users:** City Council policy analysts, RTFH outreach coordinators, and DSDP operations directors.

SYNTHESIS & NEXT STEPS

Executive Synthesis, Data Governance & Partnership Next Steps

1. EVIDENCE-BASED RESOURCE ALLOCATION

Move Beyond Seasonal Budgeting

Discontinue volatile seasonal budget adjustments. The empirical finding that 8 of 9 series exhibit zero seasonality (Overall series H=2.66, p=0.995) mandates locking investments into permanent, year-round shelter capacity and focused outreach on the top 10 persistent South East Village blocks.

2. REGIONAL ACCOUNTABILITY & SYSTEMS REFORM

Acknowledge the Regional Divergence Reality

Downtown reductions coincided with countywide unsheltered highs (6,110 in 2024). Solving unsheltered homelessness requires coordinated countywide shelter investments and inter-jurisdictional cooperation rather than localized displacement.

3. HIGH-VALUE DATA INVESTMENTS

Advance Block Digitization (R8) & Annual Retraining

Prioritize digitizing the remaining ~78 monthly block maps to uncover post-ordinance spatial shifts. Retrain the Policy Regime OLS model annually upon RTFH January PIT release using calibrated Student's t prediction intervals.